BICOL UNIVERSITY
COLLEGE OF ENGINEERING

DEPARTMENT OF CHEMICAL ENGINEERING

CHEM 13 - ORGANIC CHEMISTRY
ASYNCHRONOUS ACTIVITY 2

ANDREY S. RICAHUERTA

BS CHEMICAL ENGINEERING 2A

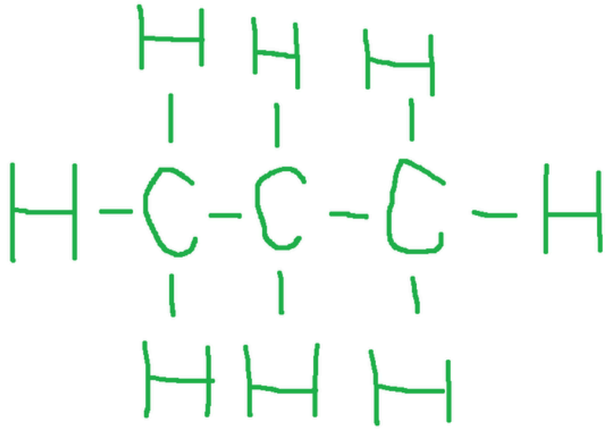
CARD 01 - Propane

Molecular Formula: C_3H_8

1. Molecular Formula

C_3H_8

2. Lewis Structure



3. Condensed Structural Formula

$CH_3CH_2CH_3$

4. Bond-Line Formula (Ketcher)



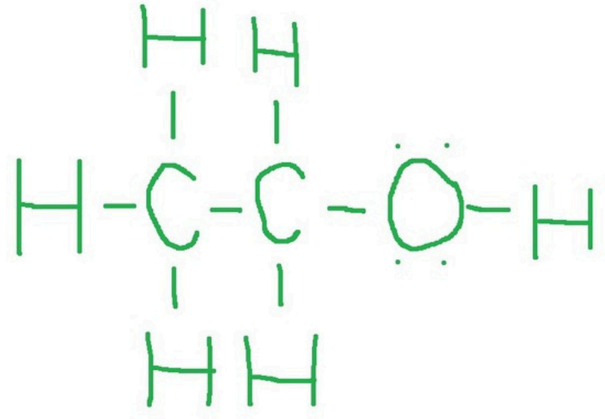
atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

CARD 02 - Ethanol
Molecular Formula: C_2H_6O

1. Molecular Formula

$$C_2H_6O$$

2. Lewis Structure



3. Condensed Structural Formula

$$CH_3CH_2OH$$

4. Bond-Line Formula (Ketcher)



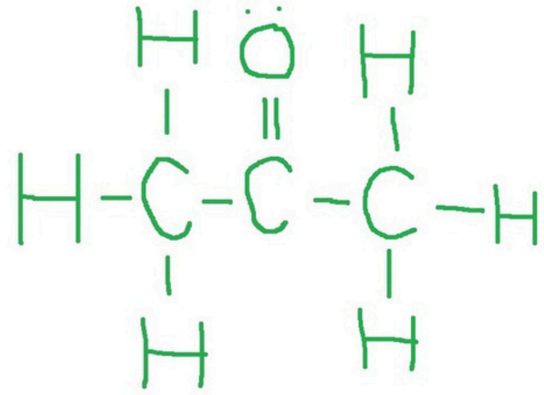
atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

CARD 03 - Acetone (propan-2-one)
Molecular Formula: C_3H_6O

1. Molecular Formula

$$C_3H_6O$$

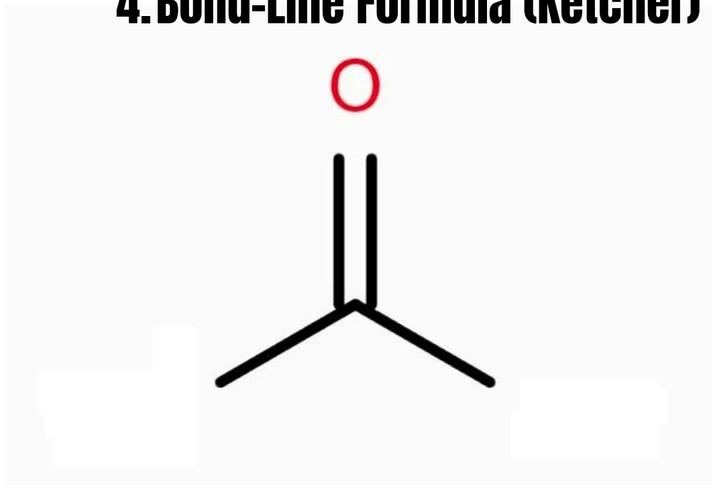
2. Lewis Structure



3. Condensed Structural Formula

$$CH_3COCH_3$$

4. Bond-Line Formula (Ketcher)



atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

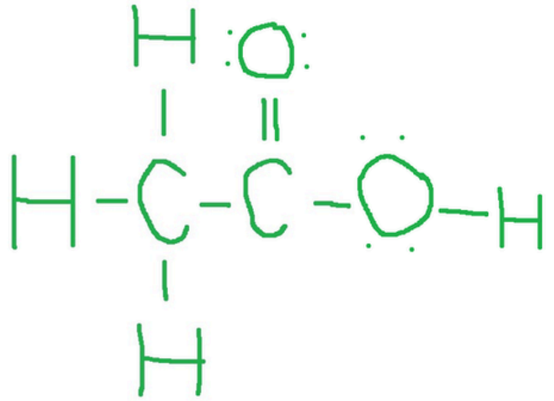
CARD 04 - Acetic acid (ethanoic acid)

Molecular Formula: $C_2H_4O_2$

1. Molecular Formula

$$C_2H_4O_2$$

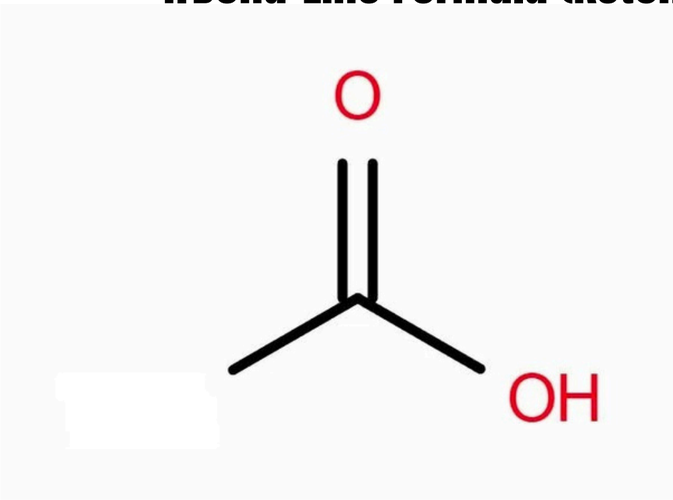
2. Lewis Structure



3. Condensed Structural Formula

$$CH_3COOH$$

4. Bond-Line Formula (Ketcher)



atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

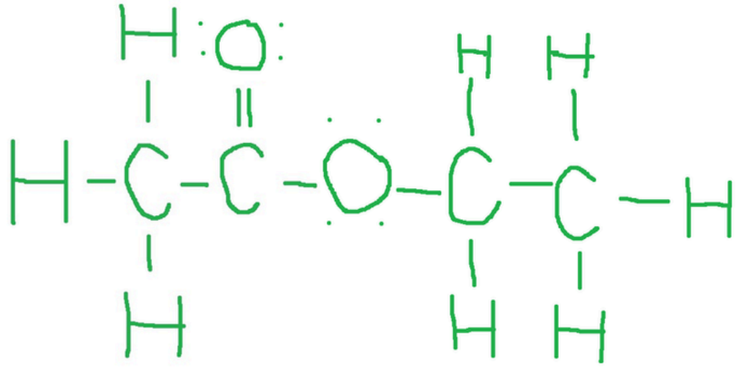
CARD 05 - Ethyl Acetate (ethyl ethanoate)

Molecular Formula: $C_4H_8O_2$

1. Molecular Formula

$$C_4H_8O_2$$

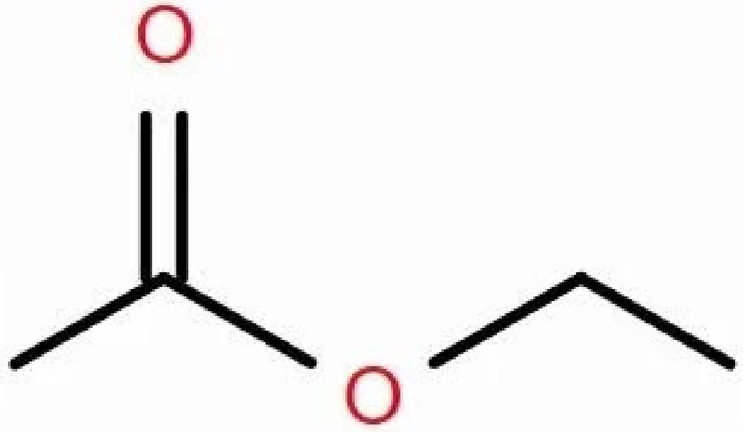
2. Lewis Structure



3. Condensed Structural Formula

$$CH_3COOCH_2CH_3$$

4. Bond-Line Formula (Ketcher)



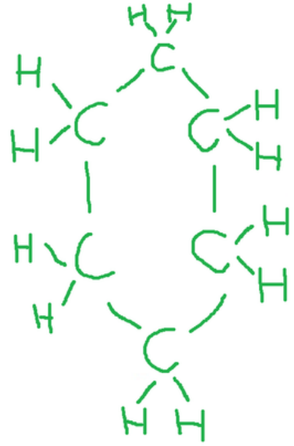
atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

CARD 06 - Cyclohexane
Molecular Formula: C_6H_{12}

1. Molecular Formula

$$C_6H_{12}$$

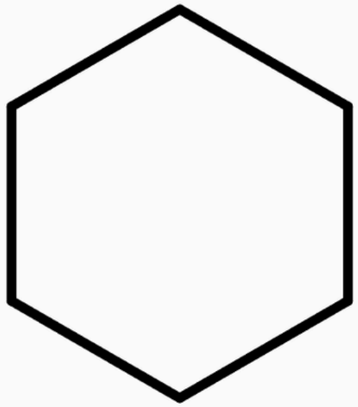
2. Lewis Structure



3. Condensed Structural Formula

$$(CH_2)_6$$

4. Bond-Line Formula (Ketcher)



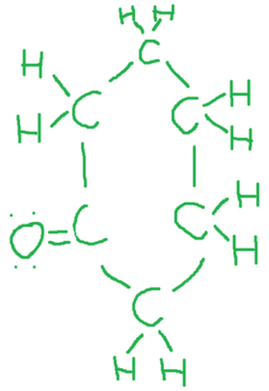
atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

CARD 07 - Cyclohexanone
Molecular Formula: $C_6H_{10}O$

1. Molecular Formula

$$C_6H_{10}O$$

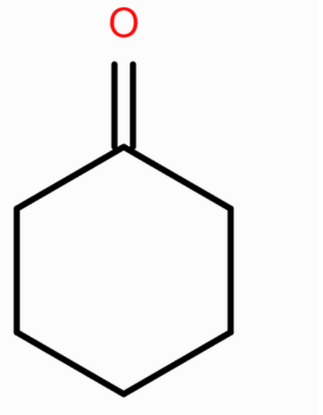
2. Lewis Structure



3. Condensed Structural Formula

$$(CH_2)_5CO$$

4. Bond-Line Formula (Ketcher)



atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

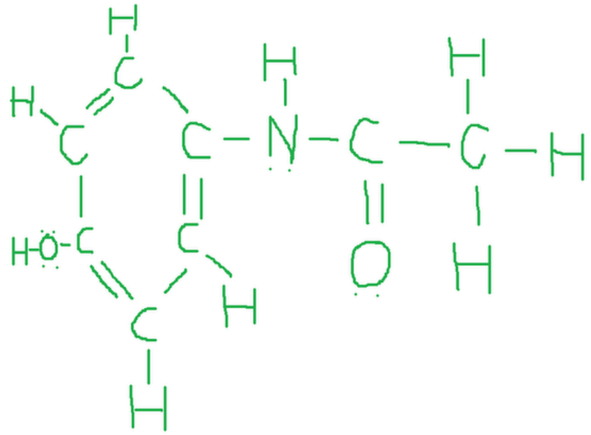
CARD 08 - Acetaminophen (paracetamol)

Molecular Formula: $C_8H_9NO_2$

1. Molecular Formula

$$C_8H_9NO_2$$

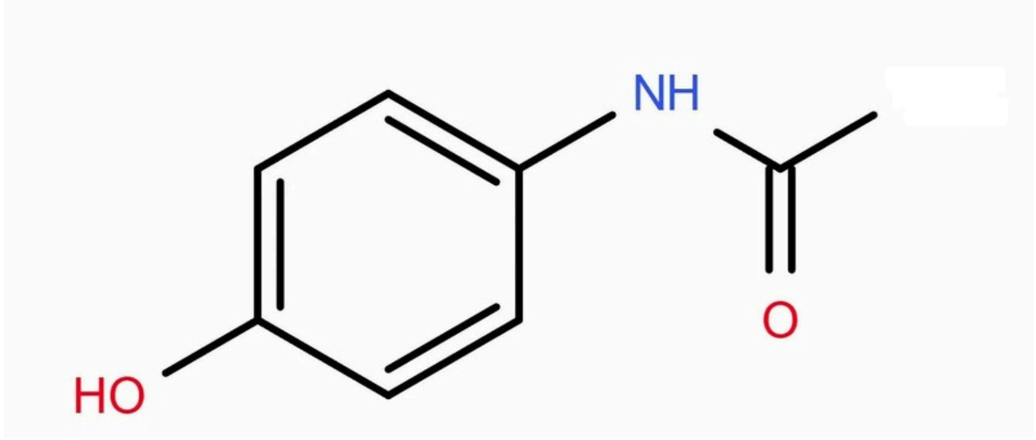
2. Lewis Structure



3. Condensed Structural Formula

$$HOC_6H_4NHCOCH_3$$

4. Bond-Line Formula (Ketcher)



atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

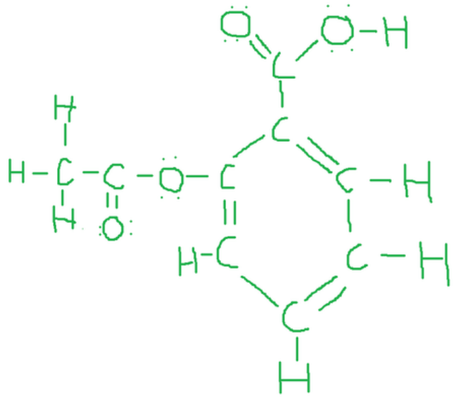
CARD 09 - Aspirin (acetylsalacylic acid)

Molecular Formula: $C_9H_8O_4$

1. Molecular Formula

$$C_9H_8O_4$$

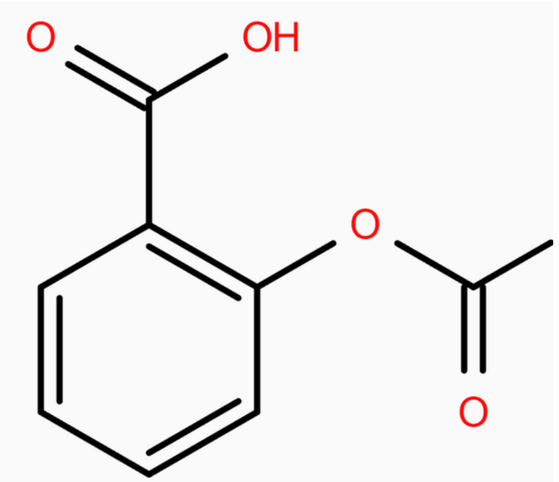
2. Lewis Structure



3. Condensed Structural Formula

$$CH_3COOC_6H_4COOH$$

4. Bond-Line Formula (Ketcher)



atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑

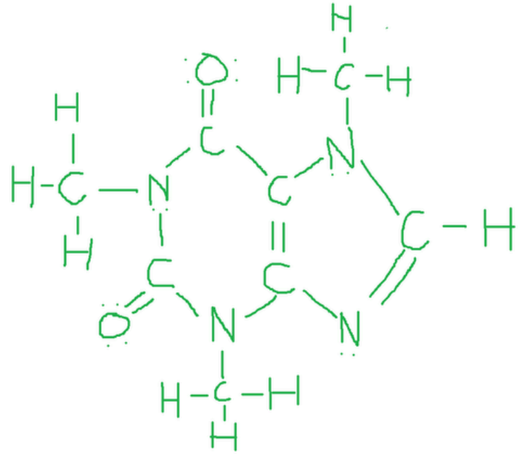
CARD 10 - Caffeine

Molecular Formula: $C_8H_{10}N_4O_2$

1. Molecular Formula

$$C_8H_{10}N_4O_2$$

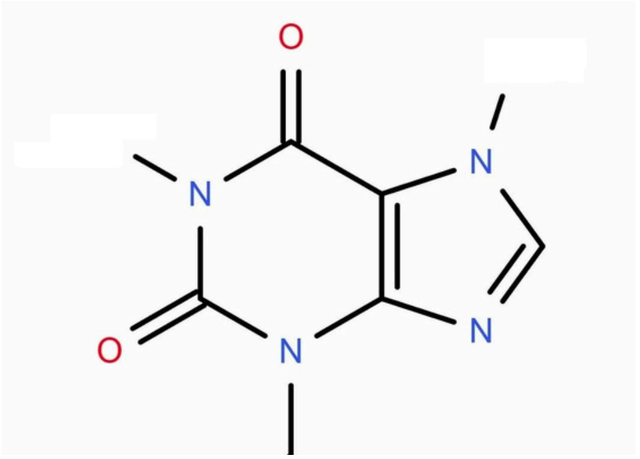
2. Lewis Structure



3. Condensed Structural Formula

$$CH_3COOCH_2CH_3$$

4. Bond-Line Formula (Ketcher)



atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity ☑